# Woocheol Shin

Personal Website | Google Scholar | woofe@kaist.ac.kr

## Research Interest

I am broadly interested in building generative policies for sequential decision-making toward human-desired goals in the real world. My current research focuses on steering large generative models, especially diffusion and flow-based models, toward desired outputs using reinforcement learning and reward-guided optimization.

## Education

**KAIST**
*M.S. Graduate School of Data Science* — *Feb. 2025 ∼ Present*

**UNIST**
*B.S. Computer Science and Engineering* — *Feb. 2019 ∼ Feb. 2025*

## Research Experience

- **System Intelligence Lab, KAIST** — Oct. 2024 ∼ Feb. 2025
  Undergraduate Research Intern
  Advisor : Prof. Jinkyoo Park(jinkyoo.park@kaist.ac.kr)
  Focus : Generative model alignment, Diffusion/Flow models, Reinforcement learning

- **Data Mining Lab, UNIST** — Dec. 2022 ∼ Jun. 2024
  Undergraduate Researcher, Seminar Organizer
  Advisor : Prof. Junghoon Kim (junghoon.kim@unist.ac.kr)
  Focus : Network stability, Cohesive subgraph discovery

## Research Papers

*Equal contribution

- **Automated Kernel Discovery Towards Understanding High-dimensional Bayesian Optimization**
  Taeyoung Yun*, **Woocheol Shin***, Inhyuck Song, Jaewoo Lee, Jinkyoo Park
  NeurIPS 2026

- **Aligning Few-Step Generative Models by Amortizing Sample-based Variational Inference**
  Jaewoo Lee*, Hyeongyu Kang*, Dohyun Kim, Kyuil Sim, **Woocheol Shin**, Minsu Kim, Taeyoung Yun, Jeongjae Lee, Sanghyeok Choi, Tabitha Edith Lee, Jongchul Ye, Jinkyoo Park
  NeurIPS 2026

- **Diffusion Fine-Tuning via Reparameterized Policy Gradient of the Soft Q-Function**
  Hyeongyu Kang*, Jaewoo Lee*, **Woocheol Shin***, Kiyoung Om, Jinkyoo Park
  ICLR 2026

- **Diffusion Alignment as Variational Expectation-Maximization**
  Jaewoo Lee, Minsu Kim, Sanghyeok Choi, Inhyuck Song, Sujin Yun, Hyeongyu Kang, **Woocheol Shin**, Taeyoung Yun, Kiyoung Om, Jinkyoo Park
  ICLR 2026

- **Urban Traffic Network Layout Optimization with Guided Discrete Diffusion Models**
  Taeyoung Yun, Inhyuck Song, **Woocheol Shin**, Yujin Shin, Sungpil Woo, Sunghwan Lim, Jinkyoo Park
  WSDM 2026

- **Beyond Trivial Edges: A Fractional Approach to Cohesive Subgraph Detection in Hypergraphs**
  Hyewon Kim, **Woocheol Shin**, Dahee Kim, Junghoon Kim, Sungsu Lim, Hyunji Jeong
  Knowledge-Based Systems 2025

## Scholarship

- **Master's Excellence Scholarship in Science & Engineering, Korea Student Aid Foundation** 2026
- **KAIST Scholarship, KAIST** Whole semesters
- **Academic Performance Scholarship, UNIST** Whole semesters

## Awards and Achievements

- **ISE & GSDS Research Day Poster Competition, KAIST** Nov. 2025
  Excellence Award
  For the poster *Diffusion Finetuning via Reparameterized Policy Gradient of the Soft Q-function*
- **Dean's List, UNIST CSE** Sep. 2024
  Awarded to top 4 students out of 224
- **Superior semester grades, UNIST**
  Spring 2023 (4.3/4.3), Fall 2023 (4.24/4.3), Spring 2024 (4.3/4.3)

## Patent

- **Data Analysis System and Method for Detecting Cohesive Subgraph in Hypergraphs** Korean Patent
  Application, Aug. 2025
  - Application No. 10-2025-0107601
  - Applicants: UNIST; Industry-Academic Cooperation Foundation, Kongju National University
  - Inventors: Junghoon Kim, Hyewon Kim, Woocheol Shin, Dahee Kim, Hyunji Jeong

## Teaching Experience

- **Teaching Assistant, KAIST** Sep. 2025 – Dec. 2025
  - IE437: Data-Driven Decision Making and Control

## Project

- **UIRP, UNIST** Mar. 2024 ∼ May. 2024
  UNIST Undergraduate Interdisciplinary Research Project
  Topic : Generalised Anchoring Methods for Maximising User Engagement in Complex Social Networks
- **Creative Independent Project, UNIST AIGS** Oct. 2023 ∼ May. 2024
  Topic : A Budget Minimisation to Reinforcing Social Engagement
- **AICP, UNIST** Apr. 2023 ∼ Dec. 2023
  UNIST AI Challengers Program
  Topic : Bipartite Cohesive Subgraph Discovery in Networks

## Industrial Projects

- **Collaborative Project with ETRI** Feb. 2025 ∼ Present
  Topic : Reinforcement Learning-based Job Scheduling Policy Learning

## Other Experiences

- **NAVER CONNECT 1784 School Revolution Program, NAVER** Dec. 2022 ∼ Feb. 2023
  Teaching AI, SW, and Autonomous Vehicles
- **ARAMCO Coding school, ARAMCO** Aug. 2022 ∼ Nov. 2022
  Teaching Programming, IOS app development
- **Education donation activities by university students, KOFAC** May. 2019 ∼ Jul. 2019
  Teaching Programming

## References


- **Associate Professor, Jinkyoo Park**
  Department of Industrial and Systems Engineering
  Associate Professor (Adjunct), Graduate School of Data Science
  Associate Professor (Adjunct), Graduate School of AI
  KAIST
  Email: jinkyoo.park@kaist.ac.kr

- **Assistant Professor, Junghoon Kim**
  Computer Science and Engineering
  UNIST
  Email: junghoon.kim@unist.ac.kr